

easily processed format, such as JSON or CSV. Most apps that use external data sources, such as apps that integrate with social media sites, rely on API calls.

Git and GitHub

We'll base our visualization on information from GitHub (<https://github.com>), a site that allows programmers to collaborate on coding projects. We'll use GitHub's API to request information about Python projects on the site, and then generate an interactive visualization of the relative popularity of these projects using Plotly.

GitHub takes its name from Git, a distributed version control system. Git helps people manage their work on a project in a way that prevents changes made by one person from interfering with changes other people are making. When you implement a new feature in a project, Git tracks the changes you make to each file. When your new code works, you *commit* the changes you've made, and Git records the new state of your project. If you make a mistake and want to revert your changes, you can easily return to any previously working state. (To learn more about version control using Git, see Appendix D.) Projects on GitHub are stored in *repositories*, which contain everything associated with the project: its code, information on its collaborators, any issues or bug reports, and so on.

When users on GitHub like a project, they can “star” it to show their support and keep track of projects they might want to use. In this chapter, we'll write a program to automatically download information about the most-starred Python projects on GitHub, and then we'll create an informative visualization of these projects.

Requesting Data Using an API Call

GitHub's API lets you request a wide range of information through API calls. To see what an API call looks like, enter the following into your browser's address bar and press ENTER:

```
https://api.github.com/search/repositories?q=language:python+sort:stars
```

This call returns the number of Python projects currently hosted on GitHub, as well as information about the most popular Python repositories. Let's examine the call. The first part, <https://api.github.com/>, directs the request to the part of GitHub that responds to API calls. The next part, `search/repositories`, tells the API to conduct a search through all the repositories on GitHub.

The question mark after `repositories` signals that we're about to pass an argument. The `q` stands for *query*, and the equal sign (`=`) lets us begin specifying a query (`q=`). By using `language:python`, we indicate that we want information only on repositories that have Python as the primary language. The final part, `+sort:stars`, sorts the projects by the number of stars they've been given.